

APSMO 2012 Set 5

1 Problems

Problem 1.1 In the multiplication problem, A, B, C represents different digits. If $\times$ means times, what digit does C represent?

$$\begin{array}{r} AB \\ \times A \\ \hline CCC \end{array}$$

Problem 1.2 The product of three numbers is 140.

If the sum of the three numbers is between 16 and 21, what is the greatest of those three numbers?

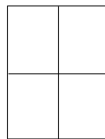
Problem 1.3 A basketball team scored 95 points in one game. They made twice as many field goals as free throws.

If each field goal scores 2 points and each free throw scores 1 point, how many field goals did the team make?

Problem 1.4 The perimeter of a rectangular piece of paper is 50cm.

It is cut into 4 congruent rectangles as shown.

What is the total of the perimeters of the four smaller rectangles?



Problem 1.5 A number bigger than 2 and less than 100 has the following property: if it is divided by 2, 3, 4, or 5, the remainder is 1 in each case.

What is the number?